rasika khante cs23172 Aim: To implement Linear Regression for predicting continuous target variables and evaluate the model performance using appropriate metrics.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score, mean_absolute_error
```

```
housing =fetch_california_housing()

df = pd.DataFrame(housing.data, columns=housing.feature_names)
df['target'] = housing.target

print(df.head())
```

```
df.shape
```

```
df.describe()
```

```
df.info()
```

```
x= df.drop('target', axis=1)
y= df['target']
```

```
print(x)
```

```
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size=0.2, random_state=42)
```

```
x_train.shape
```

```
x_test.shape
```

```
y_train.shape
```

```
y_test.shape
```

```
model= LinearRegression()
model.fit(x_train, y_train)
```

```
y_pred =model.predict(x_test)
```

```
mae=mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print("\nModel Evalution metrics")
print("----------------------------------------")
print(f"Mean Absolute Error (MAE): {mae}")
print(f"Mean Squared Error (MSE): {mse}")
print(f"Root Mean Squared Error (RMSE): {rmse}")
print(f"R-squared (R²): {r2}")
```

```
plt.scatter(x_test['MedInc'], y_test)
plt.scatter(x_test['MedInc'], y_pred)
plt.xlabel('MedInc income')
plt.ylabel('House Value')
plt.title('Linear Regression on California Hoising Dataset')
plt.show()
```

Start coding or generate with AI.